

Irreducibility of an arithmetic-progression polynomial

Theorem 1. *For every integer $n \geq 2$, the polynomial*

$$f_n(x) = \sum_{k=1}^n kx^{n-k} = x^{n-1} + 2x^{n-2} + \cdots + (n-1)x + n$$

is irreducible in $\mathbb{Q}[x]$ and in $\mathbb{Z}[x]$.

The proof has two stages. First, we compare local resultant bounds with complex root estimates and rule out factorizations whose two constant terms are at least 13. Such a factorization would have $n \geq 182$. The final section excludes smaller constant terms and completes the proof, including $2 \leq n < 182$. The condition on the constant terms is essential in the first stage; it does not follow from $n \geq 182$ alone.

Throughout, $N = n + 1$, $\ln$ denotes the natural logarithm, and $\text{rad}(m)$ is the product of the distinct prime divisors of a positive integer m .

1 Root estimates

Direct multiplication gives

$$(x-1)f_n(x) = \sum_{j=1}^n x^j - n, \quad F(x) := (x-1)^2 f_n(x) = x^N - Nx + n. \quad (1)$$

Also $f_n(1) = nN/2$, so the root 1 of F has multiplicity exactly two.

Lemma 2 (The root annulus). *Every root α of f_n satisfies*

$$1 < |\alpha| < (2n+1)^{1/n}.$$

Proof. If $|\alpha| \leq 1$, the first identity in (1) gives

$$n = \left| \sum_{j=1}^n \alpha^j \right| \leq \sum_{j=1}^n |\alpha|^j \leq n.$$

Equality forces $\alpha = 1$, which is not a root of f_n . The second identity gives $\alpha^n = N - n/\alpha$, and therefore $|\alpha|^n \leq N + n/|\alpha| < 2n + 1$. $\square$

Lemma 3 (An inner bound). *If $n \geq 3$, every root α satisfies $|\alpha|^n > 4$. For an upper-half-plane root $\alpha = \rho e^{i\theta}$, with $0 < \theta < \pi$, one also has $\theta > 2\pi/n$.*

Proof. The real and imaginary parts of $N - (n/\rho)e^{-i\theta}$ are positive, so $n\theta = 2\pi k + \phi$, where $k \geq 0$ is an integer and $0 < \phi < \pi/2$. If $k = 0$, every α^j , $1 \leq j \leq n$, has positive imaginary part, contradicting $\sum_{j=1}^n \alpha^j = n$. Hence $\theta > 2\pi/n$.

When $\theta < \pi/2$, distance to the ray of argument $-\theta$ gives

$$\rho^n \geq N \sin \theta > 2N\theta/\pi > 4N/n > 4.$$

When $\theta \geq \pi/2$, the same root identity gives $\rho^n > N$. There are no positive real roots, since all coefficients are positive. A negative real root satisfies $\rho^n = N + n/\rho > N \geq 4$. Conjugation handles the lower half-plane. $\square$

Lemma 4 (The sharper inner bound). *If $n \geq 8$, every root α satisfies*

$$n \ln |\alpha| > 11/6.$$

Proof. In the acute-angle case of Lemma 3, $|\alpha|^n > N \sin(2\pi/n) > 2\pi$. The last inequality follows from $\sin t \geq t - t^3/6$ and $n^2/(n+1) \geq 64/9 > 2\pi^2/3$. The other cases give $|\alpha|^n > N \geq 9 > 2\pi$. Finally $2\pi > e^{11/6}$, so taking logarithms proves the claim. $\square$

2 Factors, coefficients, and the discriminant identity

Suppose that $f_n = gh$, where $g, h \in \mathbb{Z}[x]$ are monic and nonconstant. Until a smaller-degree factor is explicitly chosen, either factor may be called g . Write

$$d = \deg g, \quad a = g(0), \quad b = h(0), \quad g_j = [x^j]g, \quad h_j = [x^j]h.$$

All real roots of the factors are negative. Each conjugate pair contributes $|t - \alpha|^2 > 0$ to evaluation at a real $t \geq 0$. Consequently $g(t), h(t) > 0$ for $t \geq 0$; in particular $a, b, g(1), h(1)$ are positive integers. Lemma 2 gives $a, b > 1$.

Lemma 5 (Coprime constant terms). *Every such factorization satisfies*

$$ab = n, \quad \gcd(a, b) = \gcd(a, g_1) = 1. \quad (2)$$

In particular, $n \geq 6$.

Proof. Compare the constant and linear coefficients: $ab = n$ and $ah_1 + bg_1 = n - 1$. A common divisor of a, b , or of a, g_1 , would divide both n and $n - 1$. Since $a, b > 1$ are coprime, $ab \geq 2 \cdot 3$. $\square$

Define the integer

$$J = g_1 - ag_{d-1}.$$

Lemma 6 (The coefficient difference). *If $\alpha_1, \dots, \alpha_d$ are the roots of g , then*

$$J = a \sum_{i=1}^d (\alpha_i - \alpha_i^{-1}), \quad \gcd(a, J) = 1.$$

In particular $J \neq 0$.

Proof. The root formulas for the two coefficients are $g_{d-1} = -\sum_i \alpha_i$ and $g_1 = -a \sum_i \alpha_i^{-1}$. Also $J \equiv g_1 \pmod{a}$, so Lemma 5 gives $\gcd(a, J) = 1$. Since $a > 1$, this excludes $J = 0$. $\square$

Lemma 7 (The size of the coefficient difference). *Put $L = \ln(2n+1)$ and $K_n = 2 \sinh(L/n)/(L/n)$. Then*

$$0 < |J| \leq K_n a \ln a. \quad (3)$$

Moreover, $K_n < 21/10$ for $n \geq 6$, and $K_n < 201/100$ for $n \geq 1000$.

Proof. Conjugation permutes the roots, so Lemma 6 gives

$$|J| = a \left| \sum_i (\alpha_i - \bar{\alpha}_i^{-1}) \right| \leq a \sum_i 2 \sinh(\ln |\alpha_i|).$$

Use $0 < \ln |\alpha_i| < L/n$, the monotonicity of $\sinh t/t$, and $\sum_i \ln |\alpha_i| = \ln a$. The quantity L/n decreases with n ; it is below $1/2$ for $n \geq 6$ and below $1/100$ for $n \geq 1000$. The inequality $\sinh t/t \leq (1 - t^2/6)^{-1}$ gives the stated constants. $\square$

Lemma 8 (The derivative at a root). *At every root α of f_n ,*

$$F'(\alpha) = \frac{nN(\alpha - 1)}{\alpha}, \quad f'_n(\alpha) = \frac{nN}{\alpha(\alpha - 1)}.$$

In particular, all roots of f_n are simple.

Proof. By (1),

$$F'(\alpha) = N(\alpha^n - 1) = N(n - n/\alpha) = \frac{nN(\alpha - 1)}{\alpha}.$$

Differentiating $F = (x - 1)^2 f_n$ at a root gives $F'(\alpha) = (\alpha - 1)^2 f'_n(\alpha)$. Lemma 2 ensures $\alpha \neq 0, 1$, so division is legitimate and the resulting derivative is nonzero. $\square$

Lemma 9 (The discriminant–resultant identity). *Every nontrivial monic integer factorization satisfies*

$$|\text{Disc}(g)| |\text{Res}(g, h)| a g(1) = (nN)^d. \quad (4)$$

Consequently, every prime divisor of $\text{Res}(g, h)$ divides nN .

Proof. Write $g(x) = \prod_{i=1}^d (x - \alpha_i)$. Multiplying the identity in Lemma 8 gives

$$\left(\prod_i f'_n(\alpha_i) \right) \left(\prod_i \alpha_i \right) \left(\prod_i (\alpha_i - 1) \right) = (nN)^d.$$

Evaluation of g at 0 and 1 shows that

$$\prod_i \alpha_i = (-1)^d a, \quad \prod_i (\alpha_i - 1) = (-1)^d g(1).$$

These two signs cancel. Since $f_n = gh$ and $g(\alpha_i) = 0$,

$$\prod_i f'_n(\alpha_i) = \left(\prod_i g'(\alpha_i) \right) \left(\prod_i h(\alpha_i) \right).$$

For monic g , the second product is $\text{Res}(g, h)$. In the first, each unordered root pair contributes $(\alpha_i - \alpha_j)(\alpha_j - \alpha_i) = -(\alpha_i - \alpha_j)^2$. Thus

$$\prod_i g'(\alpha_i) = (-1)^{d(d-1)/2} \text{Disc}(g).$$

Substitution gives the signed identity

$$(-1)^{d(d-1)/2} \text{Disc}(g) \text{Res}(g, h) a g(1) = (nN)^d.$$

Take absolute values and use $a, g(1) > 0$ to obtain (4). All its factors are nonzero integers, proving the assertion about prime divisors. $\square$

3 Local root packets and resultant bounds

For a prime $p \mid nN$, let M be the member of $\{n, N\}$ divisible by p , and write $M = p^r m$, where $p \nmid m$. Work in a finite unramified extension $E/\mathbb{Q}_p$ containing the m th roots of unity. Its valuation satisfies $v(p) = 1$ and has value group $\mathbb{Z}$. Such a field is obtained from a suitable finite residue field by lifting roots of unity with Hensel's lemma. We extend v to an algebraic closure of E .

Lemma 10 (Residue classes and the root reducing to zero). *Every root of f_n is integral and reduces either to zero or to an m th root of unity. For an m th root of unity ζ , its residue class contains p^r roots if $\zeta \neq 1$, and $p^r - 2$ if $\zeta = 1$. The zero class contains exactly one root, of valuation r , if $p \mid n$; otherwise it contains none.*

Proof. Integrality follows from monicity. Modulo p , the polynomial F is $(x^m - 1)^{p^r}$ if $p \mid N$, and $x(x^m - 1)^{p^r}$ if $p \nmid n$. The roots of $x^m - 1$ are distinct because $p \nmid m$. Removing $(x - 1)^2$ gives the stated counts for f_n . When $p \mid n$, the Newton polygon segment of F from $(0, r)$ to the unit linear coefficient at $(1, 0)$ gives valuation r for the single residue-zero root. $\square$

Lemma 11 (A denominator criterion). *Let $P \in E[x]$ be monic, $\zeta \in E$, and $e \geq 1$ an integer. If every root α of P satisfies $v(\alpha - \zeta) = 1/e$, then every monic divisor of P in $E[x]$ has degree divisible by e . In particular, P is irreducible if $\deg P = e$.*

Proof. For a monic divisor A of degree s , multiplication over its roots gives $v(A(\zeta)) = s/e$. This belongs to the value group $\mathbb{Z}$ of E , so $e \mid s$. $\square$

For the unit roots, the common coefficient calculation is

$$\begin{aligned} v_p\binom{N}{j} &= r - v_p(j), & p \mid N, \quad 1 \leq j \leq p^r, \\ v_p\binom{n+1}{j} &= r - v_p(j(j-1)), & p \mid n, \quad 2 \leq j \leq p^r + 1. \end{aligned}$$

These follow from $\binom{M-1}{k} \equiv (-1)^k \pmod{p}$ for $k < p^r$. At residue one we remove the double root by using

$$f_n(1+y) = \sum_{k=0}^{n-1} \binom{n+1}{k+2} y^k. \quad (5)$$

A *packet* below means a monic factor in $E[x]$ supplied by a negative-slope segment of the Newton polygon. Its roots have a common displacement valuation $v(\alpha - \zeta)$, the negative of that slope.

Lemma 12 (Packets at odd primes). *Suppose p is odd and ζ is an m th root of unity. The roots reducing to ζ form irreducible packets of degrees*

$$e_1 = p - 2\varepsilon, \quad e_j = p^{j-1}(p-1) \quad (2 \leq j \leq r),$$

where $\varepsilon = 1$ if $\zeta = 1$, and $\varepsilon = 0$ otherwise. Every root of packet j satisfies $v(\alpha - \zeta) = 1/e_j$.

Proof. For $\zeta \neq 1$, the coefficient valuations above give the negative-slope vertices of $F(\zeta + y)$ as

$$(0, r), (p, r-1), \dots, (p^r, 0).$$

For $\zeta = 1$, formula (5) gives instead

$$(0, r), (p-2, r-1), \dots, (p^r-2, 0).$$

The horizontal lengths are the stated e_j , and each segment drops by one, so its slope is $-1/e_j$. The Newton polygon theorem gives the packets; Lemma 11 makes each irreducible. $\square$

Lemma 13 (Binary packets away from one). *Suppose $p = 2$ and $\zeta \neq 1$ is an m th root of unity. If $r = 1$, its residue class consists of one irreducible quadratic packet, with displacement valuation $1/2$. If $r \geq 2$, it consists of a quartic packet with displacement valuation $1/2$, followed by irreducible packets of degrees 2^{j-1} and displacement valuations 2^{1-j} , for $3 \leq j \leq r$. Every factor of the quartic in $E[x]$ has even degree.*

Proof. The negative-slope part of the Newton polygon of $F(\zeta + y)$ is the lower hull of

$$(0, r), (2, r-1), \dots, (2^r, 0).$$

For $r = 1$, its only segment has length two and slope $-1/2$. For $r \geq 2$, the first three points are collinear, giving length four and slope $-1/2$; each later segment has length 2^{j-1} and slope -2^{1-j} . Apply the Newton polygon theorem and Lemma 11. The latter gives only evenness of factor degrees for the quartic, whose degree is four rather than two. $\square$

Lemma 14 (Binary packets at one). *Suppose $p = 2$. The residue-one class is empty if $r = 1$. If $r \geq 2$, it consists of irreducible packets of degrees 2^{j-1} and displacement valuations 2^{1-j} , for $2 \leq j \leq r$.*

Proof. For $r = 1$, use the count in Lemma 10. For $r \geq 2$, the binomial valuations in (5) give the negative-slope vertices

$$(0, r-1), (2, r-2), \dots, (2^r-2, 0).$$

Their lengths and slopes give the stated packets, which are irreducible by Lemma 11. $\square$

In each residue class, these packet degrees sum to the count in Lemma 10, so the lists account for all $n-1$ roots.

Lemma 15 (The local resultant bound). *Define $C_p(0) = C_p(1) = 0$ and $C_p(r) = p^{r-1} + C_p(r-2)$ for $r \geq 2$. Then*

$$v_p \text{Res}(g, h) \leq mC_p(r) - 2\lfloor r/2 \rfloor. \quad (6)$$

Proof. Different residue classes have unit root differences and contribute zero. This includes the residue-zero root, regardless of its allocation. If $r = 1$, Lemmas 12, 13 and 14 give at most one irreducible packet in each unit residue class, so the resultant valuation is zero.

For odd p and $r \geq 2$, a rational factor selects whole packets from Lemma 12. Within one residue class, packets $i < j$ contribute e_i when assigned to different factors, because every cross-difference has valuation $1/e_j$. Thus the valuation is the weight of a graph cut with edge weight e_i between $i < j$.

A maximum cut can place the last two vertices on opposite sides: if they agree, flipping the penultimate vertex gains e_{r-1} and loses at most $\sum_{i < r-1} e_i \leq e_{r-1}$. Deleting the two opposite vertices contributes $\sum_{i < r} e_i = p^{r-1}$, giving the recurrence for $C_p(r)$. At residue one the contribution is $p^{r-1} - 2$, so its bound is $C_p(r) - 2\lfloor r/2 \rfloor$.

For the binary quartic P at a nontrivial residue root, described in Lemma 13, Lemma 8 gives $v(F'(\alpha)) = r$ at its roots. Each of the $r-2$ higher packets contributes one to this valuation; the other residue classes contribute zero. Therefore $v(P'(\alpha)) = 2$, and $v(\text{Disc } P) = 8$. If P is split between g and h , both parts are quadratic. After translation their constant coefficients have valuation one and their linear coefficients have valuation at least one. Each quadratic discriminant has valuation at least two, so the internal resultant valuation is at most $(8-2-2)/2 = 2$.

Two formal vertices of weight two, with an internal edge of weight two, therefore bound every allocation of this quartic. Their edges to later packets have exactly the stated weights. The cut calculation applies to $2, 2, 4, 8, \dots$; for the packets of Lemma 14, the initial formal weight is zero. Summing over the m residue classes proves (6). $\square$

Lemma 16 (Prime divisibility of the coefficient difference). *For every hypothetical factorization,*

$$\text{rad}(bN) \mid J. \quad (7)$$

Proof. If $p \mid bN$, then $p \nmid a$, so all roots of g are units. Reduce in the residue field of a valued splitting extension. By Lemmas 12 and 13, the selected degree at each nontrivial residue root is divisible by p , including a possible quadratic part of the binary quartic. Its contribution to $\sum_i (\alpha_i - \alpha_i^{-1})$ therefore vanishes in the residue field. Each summand at residue one also vanishes. Lemma 6 now gives $v_p(J) > 0$; since J is an integer, $p \mid J$. Taking the product over the distinct primes proves the claim. $\square$

For the global estimates write

$$R = \ln |\text{Res}(g, h)|, \quad B = \frac{1}{2} \ln(nN), \quad Q = \prod_{p^2 \mid nN} p.$$

Lemma 17 (Two resultant budgets). *For every hypothetical factorization,*

$$R \leq N \sum_{p \mid Q} \frac{p \ln p}{p^2 - 1}. \quad (8)$$

For $n \geq 15$ one also has

$$R \leq \left(\frac{N}{8} - 2 \right) B + NC_8, \quad C_8 = \frac{3}{8} \ln 2 + \frac{5}{24} \ln 3 + \frac{3}{40} \ln 5 + \frac{1}{56} \ln 7 < \frac{13}{20}. \quad (9)$$

Proof. By Lemma 9, only primes dividing nN occur. The recurrence in Lemma 15 gives

$$\frac{C_p(r)}{p^r} = \sum_{j=1}^{\lfloor r/2 \rfloor} \frac{1}{p^{2j-1}} < \frac{p}{p^2 - 1}.$$

Since $C_p(1) = 0$, this proves (8).

Put $B_0 = \sum_p \lfloor v_p(nN)/2 \rfloor \ln p \leq B$. Retain the negative term in (6), and replace each $M \in \{n, N\}$ by N in its positive term. Subtracting $1/8$ from each summand in the displayed series leaves positive excess only in the first layers for $p = 2, 3, 5, 7$. Hence

$$R \leq (N/8 - 2)B_0 + NC_8 \leq (N/8 - 2)B + NC_8,$$

where the last step uses $N \geq 16$. The bounds $\ln 2 < 0.7$, $\ln 3 < 1.1$, $\ln 5 < 13/8$ and $\ln 7 < 2$ give $C_8 < 4363/6720 < 13/20$. $\square$

4 The main comparison for $n \geq 182$

Choose g of smaller degree and put

$$\ell = \ln n, \quad L = \ln(2n + 1), \quad A = \ln a, \quad \delta = d/n.$$

The annulus gives $0 < A/L < \delta < 1/2$. We first establish estimates for all such factorizations. The assumption $a, b \geq 13$ will enter only in Proposition 24.

Lemma 18 (The complex discriminant estimate). *If the logarithms of the root moduli of g lie in an interval of width at most W/n , where $W > 0$, then*

$$\ln |\text{Disc}(g)| \leq (d-1)A + \frac{Wd^2}{2n} - d \ln(W/n). \quad (10)$$

Proof. Write $\alpha_i = \rho_i u_i$, with $|u_i| = 1$, and put $\sigma = W/n$, $t = e^{-\sigma}$. For $i \neq j$,

$$\frac{|\alpha_i - \alpha_j|^2}{\rho_i \rho_j} \leq t^{-1} |1 - t u_i \overline{u_j}|^2.$$

The absolutely convergent logarithmic series gives

$$\sum_{i \neq j} \ln |1 - tu_i \overline{u_j}| = - \sum_{k \geq 1} \frac{t^k}{k} \left| \sum_i u_i^k \right|^2 - d \ln(1 - t).$$

Sum the pairwise inequality and discard the nonpositive series. Using $1 - e^{-\sigma} \geq \sigma e^{-\sigma/2}$ then gives $\ln |\text{Disc}(g)| \leq (d-1)A + \sigma d^2/2 - d \ln \sigma$. $\square$

For such a width define

$$G_W = \delta(\ell - A + \ln W) - \frac{1}{2}W\delta^2.$$

Lemma 2 permits $W = L$; Lemma 4 also permits $W = L - 11/6$ for $n \geq 8$.

Lemma 19 (The energy inequality). *For every admissible width W/n ,*

$$G_W \leq \frac{R + \ln g(1)}{n} - \delta \ln(N/n). \quad (11)$$

Proof. Taking logarithms of (4) gives $\ln |\text{Disc}(g)| + R + A + \ln g(1) = d \ln(nN)$. Insert (10), rearrange, and divide by n . $\square$

Lemma 20 (The uniform upper bound). *If $n \geq 182$, then for every admissible width,*

$$G_W < \ell/8 + 33/50. \quad (12)$$

Proof. Since $g(1)h(1) = nN/2$ and both factors are positive integers, $\ln g(1) \leq 2B - \ln 2$. In (11), this cancels the $-2B$ term of (9). Now $B < \ell + 1/(2n)$ and $C_8 < 13/20$. The error above $\ell/8 + 13/20$ is at most

$$\frac{\ell/8 + 13/20 + 1/16 + 1/(16n)}{n} < 1/100 \quad (n \geq 182),$$

proving the stated bound. $\square$

Put $K = 201/100$ and

$$S(A) = \frac{3}{20}A + \frac{1}{10} \ln(KA) + \frac{97}{100}.$$

Lemma 21 (The upper bound from prime support). *If $n \geq 1000$, then*

$$G_L < S(A) + 1/50. \quad (13)$$

Proof. The primes of Q dividing a have product at most $\sqrt{a}$; all other primes of Q divide bN . By Lemmas 16 and 7,

$$Q \leq \sqrt{a} \text{rad}(bN) \leq \sqrt{a} |J| < Ka^{3/2}A.$$

For $p \geq 11$, $p/(p^2 - 1) \leq 1/10$, while

$$\sum_{p=2,3,5,7} \left(\frac{p}{p^2 - 1} - \frac{1}{10} \right) \ln p < 97/100.$$

Thus the prime sum in (8) is less than $S(A)$. Combine (8) with (11). To bound the error from N/n and $\ln g(1)/n$, also use $Q \leq \sqrt{nN}$; that error is less than $(2.1\ell + 1)/n < 1/50$ for $n \geq 1000$. $\square$

Lemma 22 (Comparison for large parameters). *If $n \geq 1000$ and $51/20 \leq A < \delta L < L/2$, the two bounds (12) with $W = L$ and (13) are incompatible.*

Proof. Here $\ell > 6.9$, $0 < L - \ell < 0.7$ and $\ln L > 2$. The expression G_L increases with δ on the stated interval, so

$$G_L > H(A) := \frac{A(\ell + \ln L - 3A/2)}{L}.$$

The following bounds cover all possible A :

range of A	lower bound
$[51/20, 3]$	$H(A) - S(A) > 1/10$
$[3, \ell/4]$	$H(A) - S(A) > 51/200$
$[\max(3, \ell/4), L/2]$	$H(A) - \ell/8 > 59/80$.

An interval with reversed endpoints is ignored. For the first row, $(\ell + 2 - 1.5A)/(\ell + 0.7)$ increases with ℓ , so

$$H(A) > H_0(A) := \frac{A(8.9 - 1.5A)}{7.6}.$$

The function $H_0 - S$ is concave; its endpoint values exceed 0.18 and 0.1, using $\ln(2.55K) < 1.7$ and $\ln(3K) < 2$. For the second row, $H(A) > 5A/8$, and $19A/40 - \ln(KA)/10 - 0.97$ increases for $A \geq 3$ and exceeds 0.255 at three.

For the last row use concavity of H . At $A = L/2$,

$$H(A) - \ell/8 = \frac{1}{2} \ln L - \frac{3}{8}(L - \ell) > 59/80.$$

At the other endpoint, for $6.9 \leq \ell \leq 12$, a lower bound is $3(\ell - 2.5)/(\ell + 0.7) - \ell/8$. This is concave in ℓ , with endpoint values 1329/1520 and 189/254, both greater than 59/80. For $\ell \geq 12$, the lower bound is $\ell(\ell/32 + 33/80)/(\ell + 0.7) \geq 189/254$. The first two rows contradict (13); the last contradicts (12). $\square$

Lemma 23 (Comparison in the intermediate range). *If $182 \leq n < 1000$, $51/20 \leq A < \delta L < L/2$, and $W = L - 11/6$, then*

$$G_W > \ell/8 + 2/3.$$

Proof. Put $c = L - \ell$. Here $5.2 < \ell < 7$, $0.69 < c < 0.7$ and $1.4 < \ln W < 1.8$. Since G_W increases with δ ,

$$G_W > H_W(A) := \frac{A(\ell + \ln W)}{L} - \frac{A^2(L + W/2)}{L^2}.$$

For $51/20 \leq A \leq L/2$, we have $H_W(A) \geq H_W(L/2)$. Indeed,

$$H_W(A) - H_W(L/2) = \frac{A - L/2}{L} T, \quad T = \ell + \ln W - (A + L/2)(1 + W/(2L)),$$

and

$$2LT \leq P(\ell, c) := \frac{1}{2}\ell^2 - \ell c - \frac{3}{2}c^2 - \frac{47}{15}(\ell + c) + \frac{187}{40} < 0.$$

The polynomial P decreases in c , and $P(\ell, 0.69)$ increases in ℓ on this rectangle. Its maximum is

$$P(7, 0.69) = -27869/60000 < 0.$$

Consequently

$$G_W > H_W(L/2) = \ell/8 + \frac{1}{2} \ln W - \frac{3}{8}c + \frac{11}{48} > \ell/8 + \frac{2}{3}.$$

All terminating decimals in these comparisons are exact rational constants; the logarithmic bounds follow from the exponential series. $\square$

Proposition 24 (The main obstruction). *There is no nontrivial monic integer factorization $f_n = gh$ with $g(0), h(0) \geq 13$.*

Proof. If $a, b \geq 13$, Lemma 5 gives $n = ab \geq 13 \cdot 14 = 182$. Choose g of smaller degree. Then $A \geq \ln 13 > 51/20$ and $A < \delta L < L/2$. For $n \geq 1000$, Lemmas 20, 21 and 22 contradict one another. For $182 \leq n < 1000$, Lemma 23 contradicts Lemma 20, since $2/3 > 33/50$. $\square$

5 Excluding small constant terms and completing the proof

It remains to rule out factors of constant term $2, \dots, 12$. The arguments in this section apply to every $n \geq 2$, so they also settle all $n < 182$. We again allow either factor to be called g .

Lemma 25 (Parity of a factor degree). *Every hypothetical factorization satisfies*

$$d \equiv \mathbf{1}_{2|a} \pmod{2}.$$

Proof. At $p = 2$, Lemmas 13 and 14 show that every selected unit packet has even degree. By Lemma 10, the residue-zero root exists only when $2 \mid n$; it belongs to g exactly when $2 \mid a$, by taking valuations of the product of its roots. This proves the congruence. $\square$

Lemma 26 (A norm identity and lower bound). *With $t = n/N$ one has*

$$a^N = N^d g(t) > (\sqrt{nN})^d. \quad (14)$$

Proof. The integer nd is even. This is automatic if n is even; if n is odd, then N is even and $F(x) > 0$ for $x < 0$, so g has no real roots and d is even. Multiplying $\alpha^N = N(\alpha - t)$ over the roots of g now gives $a^N = N^d g(t)$, with the signs cancelling because nd is even. For $|z| > 1$,

$$|z - t|^2 - t|z - 1|^2 = (1 - t)(|z|^2 - t) > 0.$$

Multiplication gives $g(t) > t^{d/2} g(1) \geq t^{d/2}$, and hence $N^d g(t) > (\sqrt{nN})^d$. $\square$

Lemma 27 (The norm congruences). *Every hypothetical factorization satisfies*

$$a^N \equiv (-1)^d \pmod{N}, \quad a^n \equiv 1 \pmod{b}. \quad (15)$$

Proof. Expand $N^d g(n/N)$ in (14). Modulo N only its leading term n^d survives, giving the first congruence. In $(\mathbb{Z}/b\mathbb{Z})[x]/(g)$, the element x is a unit since $\gcd(a, b) = 1$. The identity $F(x) = 0$ reduces to $x^{n+1} - x = 0$, so $x^n = 1$. Multiplication by x on this free module of rank d has determinant $(-1)^d a$. Taking determinants and using the evenness of nd , proved in Lemma 26, gives the second congruence. $\square$

Lemma 28 (A cyclotomic resultant). *Suppose $N = p^r$ is a prime power and g is the smaller-degree factor of a hypothetical factorization. Then*

$$|\text{Res}(g, \Phi_N)| = p^d,$$

where Φ_N is the N th cyclotomic polynomial.

Proof. We have $0 < d < N/2 \leq \varphi(N)$ and $f_n \bmod p = (x-1)^{N-2}$. Every root α of g therefore has $v_p(\alpha-1) \geq 1/d > 1/\varphi(N)$: its ramification index is at most its local field degree, which is at most d . By Eisenstein's criterion for $\Phi_N(x+1)$, a primitive N th root ζ satisfies $v_p(\zeta-1) = 1/\varphi(N)$. Thus $v_p(\Phi_N(\alpha)) = 1$ and the resultant has valuation d .

Also $f_n(\zeta) = N/(1-\zeta)$, whence $|\text{Res}(f_n, \Phi_N)| = N^{\varphi(N)}/p$. The resultant for g divides this integer, a pure power of p . Its absolute value is therefore p^d . $\square$

Lemma 29 (Prime-power successors). *If $n \geq 2$ and $n+1$ is a prime power, then f_n is irreducible.*

Proof. Suppose $N = p^r$ and choose g of smaller degree. Put $H = \sqrt{nN}$. For odd p and $r \geq 3$, Lemma 3 gives

$$|\Phi_N(\alpha)| \geq \frac{|\alpha|^N - 1}{|\alpha|^{N/p} + 1} > \frac{3}{8} |\alpha|^{\varphi(N)}.$$

Multiply over the roots of g and use Lemmas 28 and 26. This yields $p > \frac{3}{8} H^{1-1/p}$, whereas $H > p^3 - 1 > \frac{8}{9} p^3$ gives $H^{1-1/p} > \frac{8}{9} p^2 \geq \frac{8}{3} p$, a contradiction.

For $p = 2$, use $\Phi_N(x) = x^{N/2} + 1$ instead. The same lemmas give $2^d > a^{N/2}/2^d > (\sqrt{H}/2)^d$, hence $H < 16$. This excludes $N \geq 32$. For $N = 4, 8$ the constant n is prime. For $N = 16$, the constants would be 3, 5, contradicting $5^{15} \equiv 1 \pmod{3}$ in (15).

If $N = p^2$ with odd p , one factor has even degree and the other odd degree. By (15), their constants are respectively 1 and -1 modulo p , so they are at least $p+1$ and $p-1$. Their product $p^2 - 1$ forces equality, contradicting coprimality. Finally, if $N = p$ is odd prime, $f_n(x+1)$ is Eisenstein at p . The value $N = 2$ is excluded by $n \geq 2$. $\square$

Lemma 30 (Small constants other than eleven). *In any hypothetical factorization, a constant term below 13 must equal 11.*

Proof. The integers a, b, N are pairwise coprime. By Lemma 29, N has at least two distinct prime divisors. Lemmas 16 and 7 give

$$30 \leq \text{rad}(bN) \leq |J| < \frac{21}{10}a \ln a,$$

and the radical contains at least three primes coprime to a . This excludes $2 \leq a \leq 7$, since $\ln 7 < 2$. For $a = 8, 9, 10, 12$, the least possible products of three allowable primes are 105, 70, 231, 385; the bound $\ln a < 3$ excludes each. $\square$

Lemma 31 (Excluding constant term eleven). *No factor in a hypothetical factorization has constant term 11.*

Proof. Suppose $a = 11$. The preceding radical bound gives $\text{rad}(bN) < 70$. It is even, is coprime to 11, and has at least three prime divisors, so it is 30 or 42. If n is even, these supports force b to be a power of two and $3 \mid N$. Lemma 25 gives even d , so $11^N \equiv 1 \pmod{3}$ contradicts odd N .

Thus b is odd. For its least prime divisor q , the order of 11 modulo q divides 11: it divides $11b$ by (15), and divides $q-1$, which is coprime to b . Order 11 would give $q \geq 23$, outside the possible supports; order one forces $q = 5$. Since N is not a prime power,

$$b = 5^u, \quad N = 2^v 3^w, \quad u, v, w \geq 1.$$

Reduce $11 \cdot 5^u + 1 = 2^v 3^w$ modulo 3, 8, 5, successively, to obtain u even, $v = 2$, and $w \equiv 2 \pmod{4}$. Hence

$$(2 \cdot 3^{w/2} - 1)(2 \cdot 3^{w/2} + 1) = 11 \cdot 5^u.$$

The two left factors are coprime odd integers greater than one. They must be 11 and 5^u ; their difference two would force $5^u = 9$ or 13, which is impossible. $\square$

Proposition 32 (Completion of the arithmetic exclusions). *Every hypothetical nontrivial monic integer factorization has $a, b \geq 13$. In particular, no such factorization exists for $n < 182$.*

Proof. Apply Lemmas 30 and 31 to each factor. Then $n = ab \geq 13 \cdot 14 = 182$ by coprimality. $\square$

Proof of Theorem 1. By Gauss's lemma, a reducible monic f_n would admit a nontrivial monic integer factorization. Proposition 32 forces both constant terms to be at least 13. Proposition 24 rules out precisely such a factorization. Thus f_n is irreducible over $\mathbb{Q}$, and monicity gives irreducibility in $\mathbb{Z}[x]$ as well. $\square$